These hypotheses define Bohr's 'atomic model', whose centenary was commemorated in 2013. With these assumptions (outlandish, but simple) Bohr manages to compute the spectra of all atoms, and even to predict accurately spectra not yet observed. The experimental success of this simple model is astonishing.

Clearly, there must be some truth in these assumptions, even if they run contrary to all contemporary notions of matter and dynamics. But why are there always only just certain orbits? And what does it mean to say that electrons 'leap'?

In Bohr's institute in Copenhagen, the most brilliant young minds of the century gather to try to give order to this jumble of incomprehensible behaviours in the atomic world, and to construct a coherent theory. The research is arduous and protracted, until a young German finds the key to unlock the door of the mystery of the quantum world.

Werner Heisenberg is twenty-five years old when he writes the equations of quantum mechanics, the same age as Einstein was when he wrote his three major articles. He does so on the basis of dizzying ideas.



Figure 4.3 Werner Heisenberg.

The intuition comes to him one night in the park behind the Copenhagen Institute of Physics. The young Werner walks about pensively in the park. It is really dark there; we are in 1925. There is only an occasional streetlamp, casting dim islands of light here and there. The pools of light are separated by large expanses of darkness. Suddenly, Heisenberg sees a figure pass by. Actually, he does not see him pass: he sees him appear beneath a lamp, then disappear into the dark before reappearing beneath another lamp, and then